Prediction of Intent for Human Robot Interaction

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***Abstract*— Intent prediction for human-robot interactions is a growing concern, as robotic applications begin to leave isolated manufacturing or industrial fields and start to integrate more with public-facing roles. Existing implementations use a fusion of pose and gaze features to train intent prediction models. However, a key challenge is obtaining a sufficiently thorough dataset for model training. In this paper, we implement a pose-based intent estimation model and data ingestion pipeline that is compatible with multiple distinct data sources to increase training volume, and combine it with a separately trained gaze model to reduce the final output's sensitivity to noise. We show strong performance in each component against reference models, and also demonstrate the final voting-based decision-making process on live video inputs.**

I. INTRODUCTION

With the progress of robotics technology and an increase in service-oriented deployments, the number of interactions between humans and robots is bound to increase. In this regard, intent prediction will be important to improve the quality of these human-robot interactions. For the purpose of this report, intent prediction is considered as the ability of a robot to first identify a person, and then determine if they will or will not interact with the robot within a short subsequent time window. With this information, a robot will then know when to approach interested parties, avoid preoccupied ones, and make relevant suggestions.

Service robots could particularly benefit from this application. An example could be a guide robot in a museum or airport, which is tasked with providing guests with information. By predicting interaction intent from nearby pedestrians, such a robot could actively approach them, or make itself more accessible by preemptively switching from an idle screen to a selection menu. A robot could also adjust the height or angle of a display and controls to accommodate patrons of varying heights or those with mobility aids. Conversely, in a busy environment, it would also be important for a robot to know which humans to avoid. It would be poor customer service to interrupt busy commuters in an airport who are more focused on catching a flight than on interaction.

In our report, we have developed two distinct trained models. The first is a pose-based sequence model to predict the overall intent to interact. This process extracts 2d poses using YOLOv11, and then uses either an LSTM or Transformer to determine the likelihood of interaction in a 2-5 second future window. The second is a gaze model, which extracts facial features using Mediapipe Face Landmarker and uses a trained Multilayer Perceptron (MLP) to determine exact pitch and yaw. The separation of the models was to allow us to to increase the volume of training data available, particularly by combining datasets for the sequence intent prediction task. Predictions made by the individual models were then converted into binary signals and combined using a voting procedure to produce a recommended action.

II. RELATED WORKS

There is existing work in the field of robot-human intent prediction, most of which base their predictions on RGB video input. From this source, meaningful data can be extracted in multiple ways. Some methods fuse multiple simple feature sets, including gaze, subsets of facial landmarks, and chest position [1]. Other approaches combine full body pose extraction with an additional face mesh, or an emotion detection model [2][4]. A model or multiple models can be trained on the extracted features to predict the intent to interact. This can be done via LSTM, as the intent prediction task is typically a sequence-to-sequence or sequence-to-single task [1]. In more recent approaches, due to their effectiveness in other sequence based fields, transformer networks have also been attempted for this part of the task [4]. Another way for intent prediction is to assume that intent is solely indicated through gaze direction. Therefore, if ”mutual gaze” is achieved, if the person is looking directly at the robot, there exists intent to interact [2][3]. A benefit of this method is that it reduces the ”black box” nature of when or what a model defines as intent to interact. However, substituting interaction with gaze also introduces limitations and possible miscommunication for individuals who may be glancing around. A further contrasting approach is to extract movement trajectories. This can be done using Hidden Markov Models, and the movement trajectories themselves can be classified using transformers or LSTM[5].

III. METHODOLOGY

A. Solution Overview

Our intent prediction method proposes a fusion of several methods presented in literature, by combining a model predicting the intent to interact on extracted pose with a separate gaze extraction model. This is implemented in the form of two trained deep neural networks, both operating on the same input stream, and the combination of their outputs to predict intent. The input to the entire system consists of segments of RGB video data, which is then processed by two separate pipelines. The first pipeline begins by extracting keypoint positions from the video using YOLOv11, which are ingested by a deep learning model to produce a single binary prediction of interaction intent. Both LSTM and transformer were tested as the model for this segment. The second pipeline overlays a face mesh on the video using MediaPipe

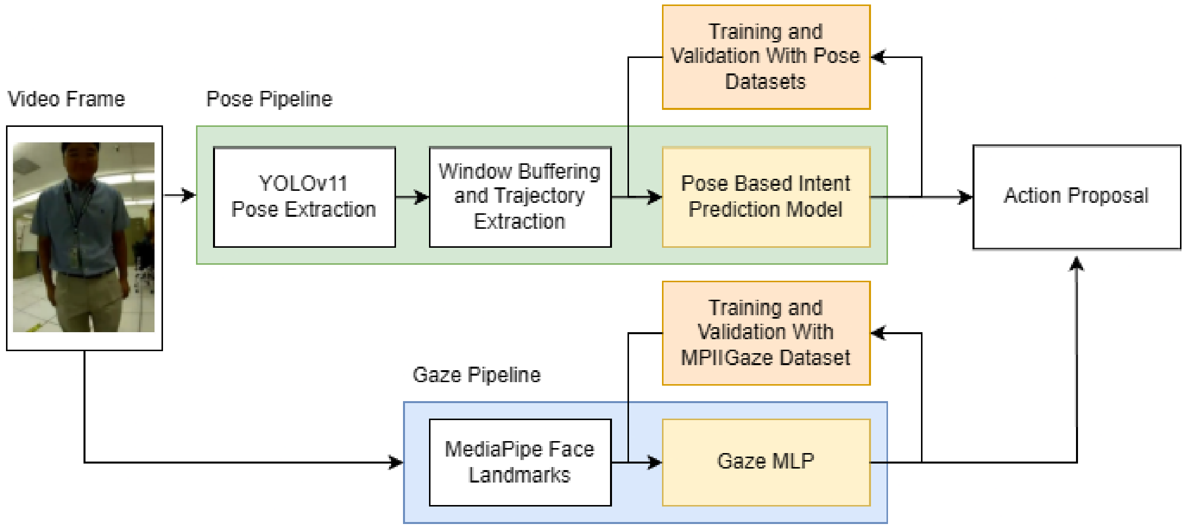

Fig. 1. Overview of Full Prediction Process with Both Trained Models

Face Landmarker. The output landmarks are then utilized by a MLP to produce an estimated gaze vector. Finally, the results of this gaze vector and the intent prediction model are combined to provide the robot with a recommended action. The action recommendation logic is presented as table I. The separation of detailed gaze extraction and intent prediction from key points is largely due to a restriction in training data, and this method was chosen to provide the largest volume of data for both models.

TABLE I
FINAL ACTION MATRIX FROM MODEL OUTPUTS

Intent Model Out	Gaze Model Out	Recommended Action
True	True	Seek Interaction
True	False	Receptive to Interaction
False	True	Receptive to Interaction
False	False	Avoid Interaction

B. Problem Definitions

Problem definitions are required for training both sub-models, as well as the overall intent prediction pipeline.

To train the intent prediction sub-model, the following definition was used. The input data space consists of 2-second sequences captured at 5Hz. At each point in time, the data consists of a 35-dimensional vector, which consists of a scaled x and y position of the 17 key points provided by YOLOv11, concatenated with a scaled scalar that indicates the rotation of the subject's face along the z axis. The ground truth targets were selected as the intent to interact, meaning they were set to true for any sequence before the end of an interaction, and false once the interaction concludes.

For the gaze sub model, the input definition would consist of face images provided by the MPIIFaceGaze dataset. Labels consist of pitch and yaw values in degrees for each static image.

The final input for the combined pipeline is a continuous video stream. Video alone was chosen as it would reduce the dependence of the overall solution on additional sensors such as LIDAR or RGB-D, making ultimate deployment more flexible. Additionally, we wanted to combine multiple ground truth datasets to improve model training performance, and some datasets did not contain depth information. The output consists of three classes as shwon in table I: for ”seek interaction” when both intent and gaze models return positive results, ”receptive to interaction” when only one model returns a positive result, and ”avoid interaction” when both models return negative results. The effectiveness of this final output was evaluated through qualitative observation of video outputs, due to unfortunate dataset limitations.

C. Dataset Availability and Pre-Processing

Four datasets were utilized for model training, and one was used for pipeline validation. The Yale Shutter Interaction dataset was the primary data used to train the intent prediction model. This data was collected using Xbox Kinect sensors to track and extract motion data and keypoints in 3d space. For this dataset, a robot with text-to-speech and three buttons was placed in a public indoor environment, and sequences of passing pedestrians and interactions were both captured. This dataset assigned interaction labels to instances where a subject would press a button on the robot. The second set used for intent prediction model training was the MINT-RVAE interaction dataset. This dataset consists of YOLOv11 keypoints extracted from video sequences. These sequences comprise individuals interacting with a robotic arm in manufactured settings. This dataset contains manual labels at frames where the subject interacts with the robot. Both datasets were provided as sequences of varying length, at 5hz, containing zero or one interaction. After processing into 2s sub-sequences, we were able to extract 22,326 sequences from the Yale Shutter Interaction dataset and 2,064 sequences from the MINT-RVAE dataset. Interaction labels were assigned for any subsequence before the final interaction frame in the original sequence. Once processed, 63.98% of sub-sequences consist of pre-interaction, and 36.02% of non-interaction.

The third dataset is the MPIIGaze dataset, which consists of 37,667 face images from 15 participants, which are each labeled with their gaze yaw and pitch. The fourth dataset is the Columbia Gaze Dataset [10]. This dataset contains a further 5,880 images of 56 subjects with the ground truth labels of pitch and yaw directions which they are looking. For this data the subjects kept their heads static in 5 different poses and were told to gaze at points corresponding to different pitch and yaw.

The final dataset is the JPL interaction dataset. This dataset consists of manufactured videos of individuals performing several pre-assigned actions from a first-person view, including handshakes, hugs, waves, and non-interaction. This dataset was used as a test dataset for validating the entire pipeline. Although it does not provide underlying labels, qualitative observation of pipeline performance on this dataset provided some insight into how well the individual model performance translates to a more grounded application.

D. Deep Learning Intent Prediction Pipeline

Each image frame for intent prediction was first processed using YOLOv11's Large model. This pre-trained model achieves a mean Average Precision of 90% for 0.5 IoU and 66% for 0.5-0.95 IoU, when tested on the COCO 2017 keypoints dataset, which suggests that the model is successful at locating the general location of the keypoints, but may struggle slightly at precise keypoint localization. The output of this model consists of a list of bounding boxes and coordinates for the keypoints of each person detected in the frame. The next step was to create trajectories of individual people, using IoU analysis, as there could be multiple detections or discontinuities where people leave the frame while others enter. Therefore, continuous trajectories were obtained by calculating the IoU of each bounding box with bounding boxes on the previous frame, and assigning the detection to the intersection with the highest IoU surpassing a threshold of 0.5; otherwise, a new trajectory was initiated. The YOLO keypoints in pixels x_p, y_p were then converted into normalized camera coordinates x_n, y_n by inverse multiplication with the camera intrinsic matrix K. These image frames were resampled to 5Hz and extracted as 2s sequences for input to the model.

$$\begin{bmatrix} x_n \\ y_n \\ z_n \end{bmatrix} = K^{-1} \begin{bmatrix} x_p \\ y_p \\ 1 \end{bmatrix}$$

Additionally, a gaze factor was created to indicate if the subject was facing towards or away from the camera. This was done by creating a plane perpendicular to the line joining two shoulder keypoints, and intersecting the line's midpoint. Taking the distance of the nose keypoint to this plane x_g, and then calculating $arctan(x_g)$ provides a factor for the general direction the subject is facing. Finally, data was resampled from the video frequency of 30Hz to the model-compatible frequency of 5Hz by only retaining every 6th data frame, and a 2s window for the model to ingest was produced on a rolling basis. This overall implementation was used when validating the combined model on

Training the model requires a separate data processing step depending on the dataset. The Shutter Interaction dataset consists of sequences already segmented per person in a global reference frame in 3d space, and contains more keypoints than provided by YOLOv11. The keypoints were first filtered to the YOLOv11 subset. They were then transformed into a robot frame and finally, projected into 2d. For the original dataset data point coordinates x_g, y_g, z_g, this was done as the following:

$$\begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{C_{rg}} & \mathbf{t_{rg}} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} x_g \\ y_g \\ z_g \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x_n \\ y_n \\ 1 \end{bmatrix} = \begin{bmatrix} x_r/z_r \\ y_r/z_r \\ z_r/z_r \end{bmatrix}$$

Where C_{rg} and t_{rg} are the rotation matrix and translation vector from global to robot frame, respectively.

The MINT-RVAE dataset already consisted of YOLOv11-compatible 2d data. In this case, it could use the same input pipeline as prediction, where it is transformed into normalized camera coordinates using the camera intrinsics. Labels for both training datasets were produced by assigning any frame occurring before the final interaction of a specific trajectory with a 1 for "will interact" and other frames with 0 for "will not interact". This was done in line with the methodology in the MINT-RVAE paper [4].

Two models were tested, and both of their results will be compared in the following sections. The first model was an LSTM, which consists of 128 hidden units, followed by 30% dropout. This is then connected to a two-layer MLP with a single output to predict a binary intent to interact. The second model is a temporal transformer architecture model. It starts with an encoding layer, which encodes each frame's pose input to an increased hidden size. This is followed by a downsampling layer to reduce the impact of the much longer sequences. Then, sinusoidal positional encoding is done, followed by prepending a learned classification token and building an attention mask. This was then passed through the transformer, which has 6 attention heads and 3 transformer encoder layers with GeLU activation. Then the learned classification token is passed through layer normalization and a linear layer to provide the logits. Both models were trained using binary cross-entropy loss and the Adam optimizer.

E. Gaze Direction Pipeline

The Mediapipe face landmark detection model [9] was used to generate a facemesh, which contains 478 2-

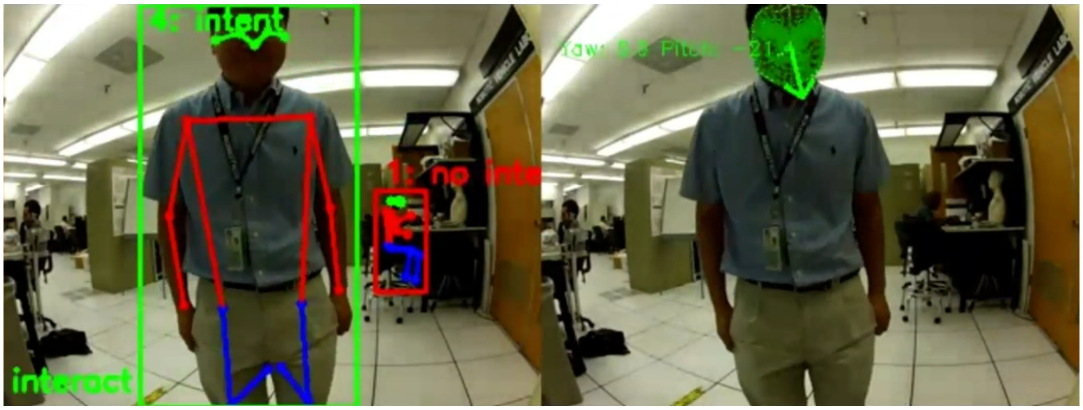

Fig. 2. Left: YOLOv11 Pose and Pose Model Interaction Intent. Right: Face Mesh Overlay and Gaze Direction. Bottom Left: Overall Combined Action Suggestion

dimensional landmarks for the person's face. These landmarks are provided with scaled coordinates between 0 and 1. An initial attempt to obtain gaze direction was to use a physics based model, based on the location of the pupil to the corners of the eye and the size of the iris. This model was based on an estimate of eyeball size, and required a clear view of the pupil and iris. While it worked when the face was close to the camera, it was ineffective at distance. Therefore, it was replaced with an MLP model that was trained using the MPIIGaze dataset. This model consists of five linear layers, with the first four followed by 1d batch normalization and 15% dropout. All layers except the final layer were activated using ReLU. It ingests the extracted face landmark keypoints, and outputs a predicted gaze direction in the form of pitch and yaw. Finally, a binary output was obtained by determining if the subject was gazing "at the robot" if it's gaze vector was within a $\pm 10°$ pitch and $\pm 10°$ yaw of the direction to the robot.

IV. RESULTS

Quantitative results are presented for both trained models in isolation, and qualitative results and observations are provided for the overall intent prediction system due to the lack of labeled ground truth.

For the LSTM and transformer trained for direct intent prediction, confusion matrices are provided on the validation dataset as Fig. 3 and Fig. 4, and show similarly strong performance for both models, with over 90% overall accuracy and relatively equal data distribution. We do notice a slight propensity for both models to predict false positives over true negatives, which could result from the slight dataset bias to positive labels. A ROC curve with AUC is also presented as figure 5. A comparison of the F1 score and AUC with reference models presented by the MINT-RVAE paper is also shown below in Table II. F1 score is compared on a sequence basis, and only the best results from each section of the reference work are tabulated.

As we can see, our model is able to outperform the reference dataset. A major factor for this performance is the inclusion of the Shutter Interaction dataset to increase

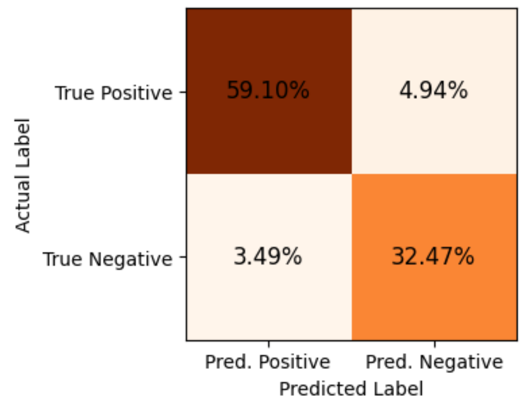

Fig. 3. LSTM Confusion Matrix

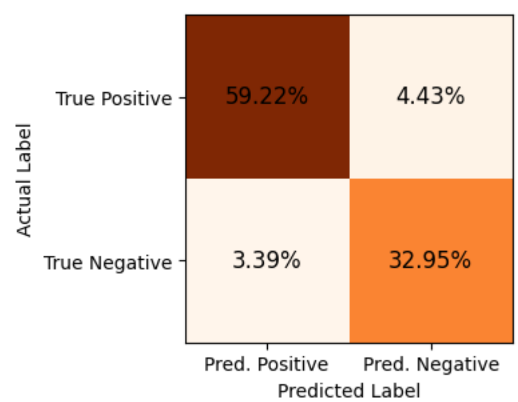

Fig. 4. Transformer Confusion Matrix

our training data volume. We were able to extract around 10x as many sequences from this dataset as from the MINT-RVAE dataset alone. This aligns with the nature of deep learning models, which generalize better when more data is provided. This also supports the significant improvement in transformer performance, as transformers are powerful models but require even larger volumes of data for effective performance.

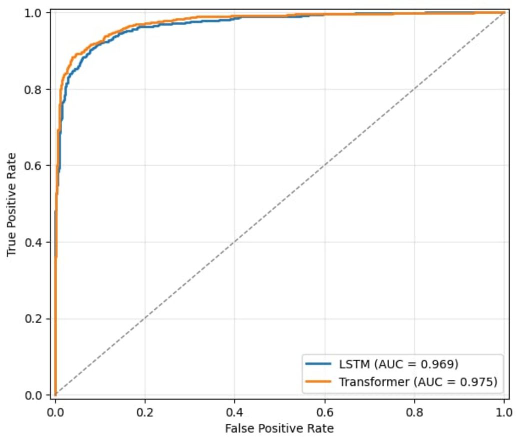

Fig. 5. AUC Curve for LSTM and Transformer

TABLE II
COMPARISON OF LSTM AND TRANSFORMER RESULTS TO MINT-RVAE REFERENCE

Source	Architecture	F1 Score	AUC
MINT-RVAE	Transformer	0.899	0.951
	LSTM	0.876	0.934
	GRU	0.876	0.933
This Paper	Transformer	0.938	0.975
	LSTM	0.933	0.969

For the gaze direction model, results are presented in table III. These results are compared with the mean error in degrees from the original MPIIGaze paper. We can observe a similar, although slightly lower performance compared to the reference paper. The similarity in performance is likely supported preprocessing performed by Mediapipe facemesh, which extracts a much smaller subset of useful features quickly and effectively. This significantly reduces the learning workload required by the downstream MLP. However, the actual gaze detection model is very simple, and ultimately used as a way to tune the final predictions, and therefore was not as refined as the reference work. These combined factors result in an overall gaze pipeline that performs similarly with reference work, but remains simple to train and implement.

TABLE III
COMPARISON OF GAZE MLP RESULTS ON MEDIAPIPE FEATURES

Source	Architecture	Mean Error (Degrees)
Pre-MPIIGaze	MnistNet	6.3
MPIIGaze	Gazenet	5.5
	Gazenet+	5.4
This Paper	Mediapipe + MLP	6.2

Overall results are obtained by combining outputs of individual models as in table I. Visual examples of these results on sequences from the JPL Interaction Datasets are presented as figure 2. From observing the full sequences, we can see that the combination of gaze and pose models

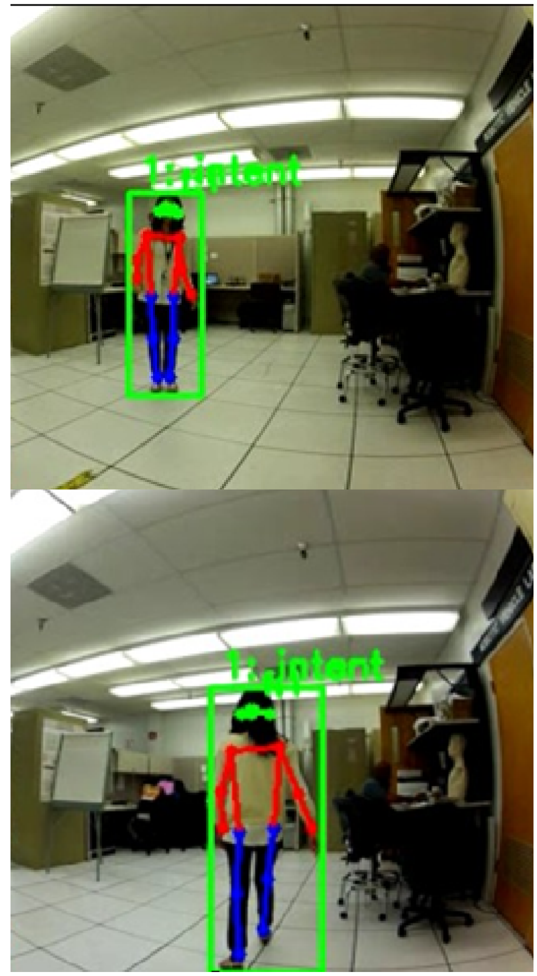

Fig. 6. Top: Proper Prediction from LSTM Model. Bottom: Improper Prediction from LSTM Model: subject is walking away from camera but YOLO provides incorrect direction assessment

reduces the noise of the final output. At times, the pose model switches from correctly predicting intent to interact to non-intent before quickly returning. This often happens as the subject approaches the frame, and is likely due to the inconsistency of YOLOv11 joint positions that are not in frame. However, when the subject is close, the gaze model is more successful in applying a face mesh and extracting gaze. Therefore, these noisy elements are suppressed by the consistent gaze in the direction of the robot, which allows the overall suggested action to switch between ”seek interaction” and ”remain receptive to interaction”. Conversely, while the face mesh has difficulty detecting faces at a distance, typically over 3 or 4 meters away, and therefore gaze cannot be obtained, the YOLOv11 joints are more easily obtained at this distance, where the subject is completely in frame. We can see here that both elements of the model are able to make up for each other's weaknesses in terms of the final output.

A problem that was observed at this final step was the the inconsistently of the subject's faced direction provided by YOLOv11. Although YOLO claims to provide an accurate

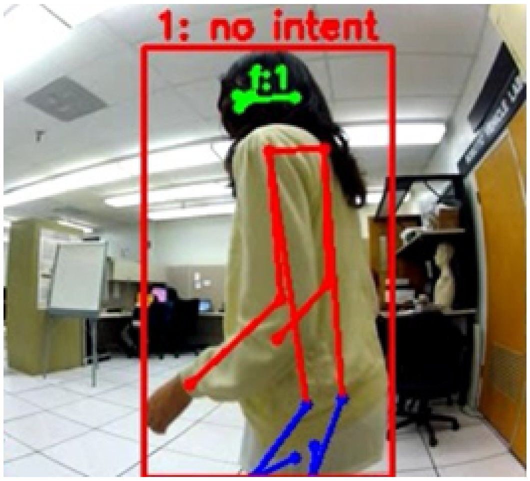

Fig. 7. Proper Intent Prediction when YOLO Nose Direction shows the Subject Facing Away

representation of which way a person is facing, using the distinction between the left and right pose joints, as well as detection of their nose, we noticed that it often produced inconsistent results, claiming that the person is facing toward the robot when they were facing away and vice versa. When this happened, it made intent prediction difficult. An example of this is seen as figure 6 where the YOLO nose pose is clearly visible even when the person is faced away. This is opposed to 7 where the YOLO derived nose position indicating the person facing away results in a successful ”non interaction” prediction. This problem could be also mitigated by extracting face direction when the gaze results are available.

V. CONCLUSIONS

The primary objective of this paper was to develop an intent prediction system that could integrate both pose and gaze features, while being robust enough to benefit from multiple distinct datasets. We trained both an LSTM and a Transformer sequence classifier on fixed-length sequences, and used 2d normalized YOLOv11 points as the input features. By using a lower dimension and the relatively smaller number of input features from YOLOv11, we were able to combine multiple datasets, increasing our training volume for both of our models. We were able to show that these models could perform more strongly than reference models trained on a larger subset of features, but on only one dataset.

We additionally delegated the gaze extraction to a separate MLP, which required pre-processing of key points using Mediapipe Face Mesh. By utilizing this pre-processing step, we reduced the training load required from the deep gaze detection model and achieved comparable performance to reference materials in real time. The outputs of both of these models were then combined in a binary fashion to determine the final suggested course of action. The entire system was integrated and validated qualitatively using the JPL interaction video dataset, where RGB video data of interactions was extracted and used for real-time interaction forecasting. This validation showed some strengths of the integration approach by reducing output noise, as well as a general indication that the model training can be successfully translated to a fully integrated system with video input.

A. Next Steps

Some additional challenges remain, as well as future developments that can be made upon the framework presented in this report. One challenge is the limitation to purely qualitative evaluation of the combined process. It would be prudent in the future to find, or manually label, several videos with interaction intention. This will allow a more thorough investigation of the effectiveness of the integrated system, and for us to determine how effectively individual model training translates to overall performance. Another next step was mentioned previously, which is the inconsistency of that YOLO has in determining when subjects are facing away from the camera.

An additional development that can be built on this framework is the prediction of specific actions. Using the already extracted face mesh and YOLOv11 poses, it would be possible to train a system to detect pre-selected actions such as waving and pointing. Additionally, actions such as these can also be detected using a deterministic algorithm based on joint angles, given that the YOLOv11 poses are provided.

REFERENCES

[1] S. Arreghini, G. Abbate, A. Giusti, and A. Paolillo, “Predicting the intention to interact with a service robot: The role of Gaze Cues,” 2024 IEEE International Conference on Robotics and Automation (ICRA), pp. 993–999, May 2024. doi:10.1109/icra57147.2024.10610289

[2] S. Arreghini, G. Abbate, A. Giusti, and A. Paolillo, “A long-range mutual gaze detector for Hri,” Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction, pp. 870–874, Mar. 2024. doi:10.1145/3610977.3637470

[3] A. Saran, S. Majumdar, E. S. Short, A. Thomaz, and S. Niekum, “Human gaze following for human-robot interaction,” 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 8615–8621, Oct. 2018. doi:10.1109/iros.2018.8593580

[4] F. Mohsen and A. Safa, MINT-RVAE: Multi-Cues Intention Prediction of Human-Robot Interaction using Human Pose and Emotion Information from RGB-only Camera Data, Sep. 2025. doi:10.48550/arXiv.2509.22573

[5] X. Zhang, S. Tian, X. Liang, M. Zheng, and S. Behdad, “Early prediction of human intention for human-robot collaboration using Transformer Network,” Volume 2: 43rd Computers and Information in Engineering Conference (CIE), Aug. 2023. doi:10.1115/detc2023-116492

[6] X. Zhang, Y. Sugano, M. Fritz, and A. Bulling, “MPIIGaze: Real-world dataset and deep appearance-based gaze estimation,” IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 41, no. 1, pp. 162–175, Jan. 2019. doi:10.1109/tpami.2017.2778103

[7] Thompson, Sydney; Alexander Lew; Rohan Phanse; Alex Huang; Elizabeth Stanish; Yifan Li; Marynel Vázquez, 2025, ”Shutter Interaction Dataset - Zipped”, https://doi.org/10.60600/YU/KFFQPF, Yale Dataverse, V1

[8] Camillo Lugaresi, undefined., et al, ”MediaPipe: A Framework for Perceiving and Processing Reality,” in Third Workshop on Computer Vision for AR/VR at IEEE Computer Vision and Pattern Recognition (CVPR) 2019, 2019.

[9] M. S. Ryoo and L. Matthies, ”First-Person Activity Recognition: What Are They Doing to Me?”, CVPR 2013.

[10] B. A. Smith, Q. Yin, S. K. Feiner, and S. K. Nayar, “Gaze locking,” Proceedings of the 26th annual ACM symposium on User interface software and technology, pp. 271–280, Oct. 2013. doi:10.1145/2501988.2501994